MedFusion AI

AI-Assisted Personal Health Summary Report

Institutional Ref: MF-2026-6YXSSW | Accredited Automated Analysis | A4 Clinical Standard

CONFIDENTIAL REPORT
For Patient & Physician Review

PATIENT IDENTIFICATION

Patient Name : Test Patient
Patient ID : PID-148729
Email Address : test@example.com

EXAMINATION & FILE DETAILS

Study Type : Chest X-ray (Frontal Projection)
Source File : sample_bone_xray.jpg
Date & Time : 11 Oct 2026 at 04:43 PM

MOSTLY NORMAL

No significant diagnostic abnormalities detected by AI model. Correlate with clinical symptoms.

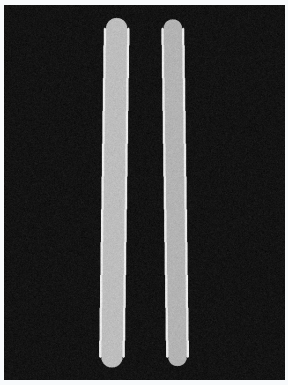
EXECUTIVE CLINICAL SUMMARY

- Bilateral lung fields demonstrate symmetric radiodensity across upper, mid, and lower zones.
- Cardiac silhouette shows prominent transverse diameter (estimated CTR 0.68).

NOTABLE OBSERVATIONS REQUIRING REVIEW:

& Borderline cardiothoracic ratio (0.68)

SOURCE IMAGE CAPTURE



Analyzed Capture: sample_bone_xray.jpg

PRELIMINARY AI CLINICAL IMPRESSION

No focal airspace consolidation, gross pleural effusion, or abnormal cardiothoracic enlargement identified by the screening model. Does not exclude sub-centimeter or subtle clinical findings.

IMPORTANT MEDICAL DISCLAIMER

MedFusion AI provides AI-assisted personal health informational analysis. It does not provide medical diagnosis, prescribe treatments, or replace a doctor. Always consult a qualified medical professional for health decisions. This is a demo report with synthetic/example data.

DETAILED CLINICAL ANALYSIS — EXAMINATION FINDINGS

Ref: MF-2026-6YXSSW | Study: Chest X-ray (Frontal Projection)

The following structured table enumerates parameters, anatomical regions, and biomarker values evaluated by the analysis pipeline.

PARAMETER / REGION	OBSERVED RESULT / MEASUREMENT	REFERENCE CRITERIA	STATUS
Lung Fields	Symmetric bilateral radiodensity	Bilateral symmetric aerated parenchyma	Within Limits
Consolidation / Opacity	No focal consolidation identified by model	No focal airspace consolidation	Within Limits
Cardiothoracic Ratio (CTR)	CTR estimated at 0.68 (Normal: <= 0.50)	<= 0.50 (Frontal projection)	Review Required
Pleural Spaces	No gross blunting or large effusion identified	Sharp, acute costophrenic angles	Within Limits
Thoracic Cage & Ribs	Visible rib contours and clavicular alignment intact	Intact cortical bone margins	Within Limits
Image Technical Quality	Optimal	Diagnostic sharpness and inspiratory effort	Adequate

TECHNICAL QUALITY METRICS & MEASUREMENT SPECIFICATIONS

Image Dimensions: 600 × 800 px (Aspect Ratio: 0.75)

Sharpness Variance (Laplacian): 1977.1 | Contrast Standard Deviation: 64.6

Mean Luminance Level: 47.5 / 255 | Overall Technical Grade: Optimal

Quality Note: No technical degradation, extreme under/over exposure, or severe motion artifacts detected.

Limitations: Automated quality scoring reflects 2D sensor capture and does not guarantee complete diagnostic visibility of microscopic features.

NUMERICAL MEASUREMENTS & ALGORITHMIC INDICES

CARDIOTHORACIC RATIO:	0.68
LOWER ZONE ASYMMETRY:	0.061
SHARPNESS VARIANCE:	1977.1

1. EVIDENCE-BASED SYNTHESIS & UNCERTAINTY ANALYSIS

No focal airspace consolidation, gross pleural effusion, or abnormal cardiothoracic enlargement identified by the screening model. Does not exclude sub-centimeter or subtle clinical findings.

Uncertainty Context: Single-view projection imaging cannot fully eliminate retrocardiac or sub-diaphragmatic overlap. Volumetric 3D features are not visible.

2. CLINICAL NEXT STEPS & FOLLOW-UP TIMELINE

1. Correlate with clinical history (cough, fever, dyspnea, auscultation findings).
2. Request formal radiologist over-read for diagnostic confirmation.
3. Repeat PA and lateral views if clinical suspicion of lower respiratory infection persists.

3. RED-FLAG WARNING SIGNS (SEEK IMMEDIATE EMERGENCY CARE)

- &¡ Severe or acute chest pain, sudden difficulty breathing, coughing up blood (hemoptysis).
- &¡ High fever (> 102°F / 38.9°C), severe dizziness, confusion, or sudden altered mental status.
- &¡ Rapidly growing, bleeding, or ulcerating skin lesion with jagged asymmetric boundaries.
- &¡ Extreme lethargy, pale clammy skin, or signs of severe dehydration / sepsis.

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VERIFIED BY ENGINE

MedFusion AI Pipeline v2.4
SHA-256 Checksum: Verified
Signed: 11 Oct 2026